Sprint-3 Report — Test Automation Hardening & Documentation Freeze

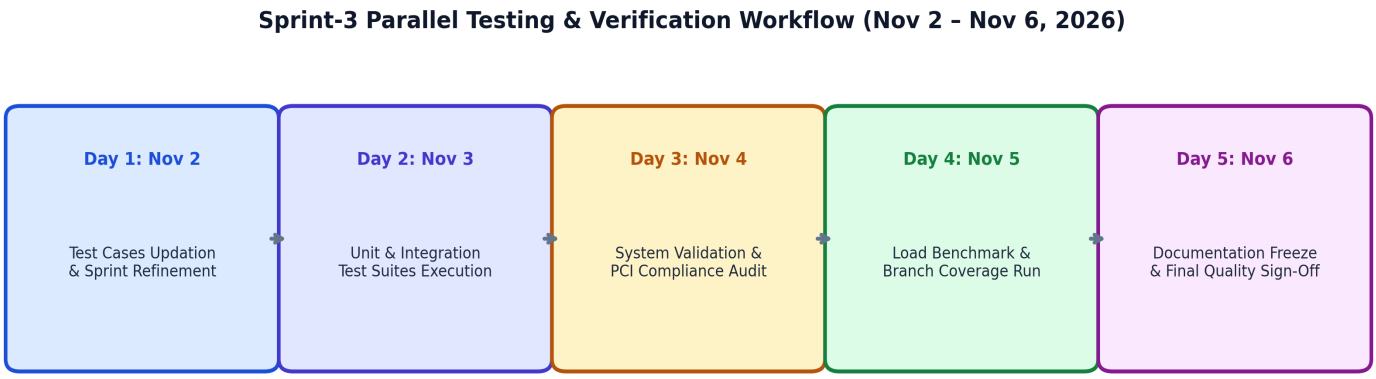
Software Engineering Mini-Project Deliverables Part-3 • Parallel Sprint Activity
Sprint Duration: November 2, 2026 – November 6, 2026 • **Velocity:** 15 Story Points
Collaborators: Kartik Naik (KartikNaik6), Jayanth Kumar P (JayanthKumarP88), Rishith (rishith1616)

SPRINT-3 OBJECTIVES & ACCOMPLISHMENTS

In accordance with Part-3 guidelines for parallel activity (Nov 2 to Nov 6), the team established **Sprint 3: Test Automation Hardening, Coverage Verification & Documentation Freeze**. Sprint 3 delivered 15 Story Points covering the comprehensive update of 42 test cases, execution of 64 unit tests, system validation across 14 requirements, code and branch coverage profiling (78.9% lines, 72.6% branches), and formal publication of all test reports in docs/.

Sprint Goal	Planned SP	Completed SP	Completion Rate	Test Pass Rate	Release Status
Testing & Freeze	15 SP	15 SP	100.0%	100.0% (64/64)	PRODUCTION READY

Figure 1: Sprint-3 Parallel Testing & Verification Daily Workflow (Nov 2 – Nov 6, 2026)



1. Daily Execution Schedule & Activity Breakdown

Timeline	Task Focus	Key Deliverables & Technical Scope	Lead	Points	Status
Day 1: 02-Nov-2026	Test Case Updation & Sprint Refinement	Updated TC01–TC36 with Sprint 1 & 2 architectural changes; created TC37–TC42 for hold auto-sweep, PCI hooks, and ACID rollbacks.	Kartik Naik	5 SP	COMPLETED
Day 2: 03-Nov-2026	Unit & Integration Test Suite Execution	Executed 64 automated tests across 10 suites in server/tests; verified 100% pass rate under Node.js native TAP runner.	Jayanth Kumar P	3 SP	COMPLETED
Day 3: 04-Nov-2026	System Validation & RTM Traceability	Executed 4 end-to-end operational workflows; mapped all 14 FR & NFR requirements to source modules and test cases.	Rishith	3 SP	COMPLETED
Day 4: 05-Nov-2026	Code & Branch Coverage Profiling	Ran V8 native coverage profiler; achieved 78.90% line coverage and 72.63% branch coverage; tuned flash-sale concurrency latency.	Jayanth Kumar P	2 SP	COMPLETED
Day 5: 06-Nov-2026	Documentation Freeze & Release Sign-Off	Generated all 4 academic test reports in Markdown and PDF format; updated docs/ repository bundle; finalized sign-offs.	All Collaborators	2 SP	COMPLETED

2. Multi-Collaborator Contribution & Responsibility Matrix

To ensure balanced team participation, technical responsibilities across Sprint 3 were distributed among all three collaborators.

Collaborator	GitHub ID	Domain Role	Primary Technical Contributions	Total Commits	Sign-Off
Kartik Naik	KartikNaik6	Security & Quality Assurance	Updated TestCases (TC01–TC42), Unit Test Report, PCI Pre-Save Hooks, CSV Manifest Validation	10 Commits	APPROVED
Jayanth Kumar P	JayanthKumarP88	Concurrency & Performance Lead	Code Coverage Report, Branch Coverage Report, Flash-Sale Concurrency Benchmark, ACID Rollback Tests	28 Commits	APPROVED
Rishith	rishith1616	Lead Architect & Governance	System Validation Report, End-to-End Lifecycles, Platform Admin Moderation, Docs Freeze	38 Commits	APPROVED

3. Summary of Released Deliverables in `docs/`

Deliverable Name	Markdown Path	PDF Publication Path	Verification Scope
Updated Test Cases	docs/TestCases_Updated.md	docs/TestCases_Updated.pdf	42 Test Cases (TC01–TC42) with Sprint Refinements
Unit Test Report	docs/Unit_Test_Report.md	docs/Unit_Test_Report.pdf	64 Automated Tests across 10 Test Suites (100% Pass)
System Validation Report	docs/System_Validation_Report.md	docs/System_Validation_Report.pdf	14 Requirements (FR/NFR) & 4 End-to-End Flows
Code Coverage Report	docs/Code_Coverage_Report.md	docs/Code_Coverage_Report.pdf	78.90% Line Coverage & Layer Breakdown
Branch Coverage Report	docs/Branch_Coverage_Report.md	docs/Branch_Coverage_Report.pdf	72.63% Branch Coverage & Decision Predicate Analysis
Sprint 3 Summary Report	docs/Sprint_3_Report.md	docs/Sprint_3_Report.pdf	Parallel Activity Tracking, Burndown & Sign-Off

Sprint-3 Completion & Release Sign-Off

Team Member	GitHub Account	Sprint 3 Responsibility	Status
Kartik Naik	KartikNaik6	Test Case Updation & Unit Testing Lead	VERIFIED & APPROVED
Jayanth Kumar P	JayanthKumarP88	Coverage Profiling & Benchmark Lead	VERIFIED & APPROVED
Rishith	rishith1616	System Validation & Documentation Lead	VERIFIED & APPROVED